

On the Theory of Modules

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§ 4. Isomorphism; the function χ .

Definition of isomorphism.

Suppose that between the residues ρ of a module $\mathfrak{a}$ and the residues ρ' of a module $\mathfrak{a}'$ one can establish a one-to-one correspondence with the following property: whenever ρ_i, ρ'_i and ρ_k, ρ'_k are two pairs of corresponding residues, then $\rho_i + \rho_k$ and $\rho'_i + \rho'_k$ also form a pair of corresponding residues. Then the correspondence is called an “isomorphic” relation between $\mathfrak{a}$ and $\mathfrak{a}'$. Two modules $\mathfrak{a}, \mathfrak{a}'$ which admit at least one isomorphic relation are called “isomorphic to one another”.¹ Thus the relation of isomorphism is always reciprocal.

If $\rho_1, \rho_2, \dots$ are all residues of $\mathfrak{a}$, and if $\rho'_1, \rho'_2, \dots$ are, in this order, the residues of $\mathfrak{a}'$ assigned to them isomorphically, then the isomorphic relation in question will be denoted by

$$\rho_i \parallel \rho'_i.$$

It follows easily from the definition of isomorphism that, in general,

$$\sum_i c_i \rho_i, \quad \sum_i c_i \rho'_i$$

are corresponding residues of $\mathfrak{a}$ and $\mathfrak{a}'$. If $\mathfrak{a}'$ is isomorphic to $\mathfrak{a}''$ and

$$\rho'_i \parallel \rho''_i$$

denotes an isomorphic relation between $\mathfrak{a}'$ and $\mathfrak{a}''$, then

$$\rho_i \parallel \rho''_i$$

is an isomorphic relation between $\mathfrak{a}$ and $\mathfrak{a}''$; it is called the relation “composed of $\rho_i \parallel \rho'_i$ and $\rho'_i \parallel \rho''_i$ ”. If one composes the isomorphic relation $\rho_i \parallel \rho'_i$ successively with all distinct isomorphic relations from $\mathfrak{a}'$ to $\mathfrak{a}''$, one obtains the same number of isomorphic relations between $\mathfrak{a}$ and $\mathfrak{a}''$, and these exhaust all of them. Thus, if

$$\chi(\mathfrak{a} \parallel \mathfrak{a}')$$

denotes the number of isomorphic relations existing between two modules $\mathfrak{a}$ and $\mathfrak{a}'$, then the following theorem holds:

If $\mathfrak{a}$ is isomorphic to $\mathfrak{a}'$, and $\mathfrak{a}'$ is isomorphic to $\mathfrak{a}''$, then $\mathfrak{a}$ is isomorphic to $\mathfrak{a}''$ and

$$\chi(\mathfrak{a} \parallel \mathfrak{a}'') = \chi(\mathfrak{a}' \parallel \mathfrak{a}'') = \chi(\mathfrak{a} \parallel \mathfrak{a}').$$

The modules $\mathfrak{a}, \mathfrak{a}', \mathfrak{a}''$ need not, of course, all be distinct. Taking $\mathfrak{a} = \mathfrak{a}''$, and assuming the isomorphism of $\mathfrak{a}$ and $\mathfrak{a}'$, one obtains

$$\chi(\mathfrak{a} \parallel \mathfrak{a}') = \chi(\mathfrak{a} \parallel \mathfrak{a}) = \chi(\mathfrak{a}' \parallel \mathfrak{a}').$$

An isomorphic relation of a module $\mathfrak{a}$ onto itself may be interpreted as a permutation among its residues. The permutations so obtained form a group; $\chi(\mathfrak{a} \parallel \mathfrak{a})$, for which we shall also simply write

$$\chi(\mathfrak{a}),$$

¹The term isomorphism is borrowed from group theory.

is the order of this group.

We now examine more closely under what conditions two modules $\mathfrak{a}$ and $\mathfrak{a}'$ are isomorphic. It is easy to see that this is so when they belong to the same class. For if $[E]$ is a unimodular transformation carrying $\mathfrak{a}$ into $\mathfrak{a}'$, then (§ 2) it also carries every residue of $\mathfrak{a}$ into a residue of $\mathfrak{a}'$ and in this way establishes a one-to-one correspondence between the residues of $\mathfrak{a}$ and those of $\mathfrak{a}'$, which is immediately recognized as isomorphic.

Conversely, let us now show that two isomorphic modules of the same degree, $\mathfrak{a}, \mathfrak{a}'$, must also belong to the same class. On the basis of the preceding considerations, and without loss of generality, we may assume that $\mathfrak{a}$ and $\mathfrak{a}'$ are the principal modules within their classes. We also exclude the case in which the norms of $\mathfrak{a}$ and $\mathfrak{a}'$ vanish, since it is irrelevant for our purpose. Let

$$\rho_i \parallel \rho'_i$$

be an isomorphic relation between $\mathfrak{a}$ and $\mathfrak{a}'$, let m be any integer, and let ρ_i, ρ'_i run successively through all residues of $\mathfrak{a}, \mathfrak{a}'$. Then the system $m\rho_i$ belongs to the module $\mathfrak{a}$ just as often as the system $m\rho'_i$ belongs to the module $\mathfrak{a}'$. The computation of the number in question is particularly simple when $\mathfrak{a}, \mathfrak{a}'$ are principal modules. If $a_1, a_2, \dots, a_n$ are the invariants of $\mathfrak{a}$, and $a'_1, a'_2, \dots, a'_n$ those of $\mathfrak{a}'$, one immediately sees that the number of residues ρ_i for which $m\rho_i$ is contained in $\mathfrak{a}$ is

$$(m, a_1)(m, a_2) \cdots (m, a_n),$$

and similarly, the number of residues ρ'_i for which $m\rho'_i$ belongs to $\mathfrak{a}'$ is

$$(m, a'_1)(m, a'_2) \cdots (m, a'_n).$$

Hence for every integer m the equality

$$(m, a_1)(m, a_2) \cdots (m, a_n) = (m, a'_1)(m, a'_2) \cdots (m, a'_n)$$

must hold; and this, as a simple consideration shows, can occur only when

$$a_1 = a'_1, \quad a_2 = a'_2, \quad \dots, \quad a_n = a'_n,$$

that is, when $\mathfrak{a}$ and $\mathfrak{a}'$ belong to the same class.

Nevertheless two modules of different degrees may certainly be isomorphic; the question of the isomorphism of two modules is settled by the following theorem.

I. Two modules are isomorphic if and only if their systems of invariants are identical after the invariants which reduce to 1 have been removed.

To prove that the stated condition is sufficient, suppose that the module $\mathfrak{a}$ has invariants

$$a_1 = 1, \dots, a_r = 1, a_{r+1}, a_{r+2}, \dots, a_n,$$

and that the module $\mathfrak{a}'$ has invariants

$$a'_1 = a_{r+1}, \quad a'_2 = a_{r+2}, \quad \dots, \quad a'_{n-r} = a_n.$$

We must show that $\mathfrak{a}$ and $\mathfrak{a}'$ are isomorphic. We may assume $\mathfrak{a}$ and $\mathfrak{a}'$ to be principal modules, and under this assumption the proof is very simple. Let

$$\rho_1, \rho_2, \dots$$

be a complete system of residues with respect to $\mathfrak{a}$, and let ρ'_i denote the row obtained from ρ_i by deleting the first r elements. Then it follows at once that

$$\rho'_1, \rho'_2, \dots$$

form a complete system of residues of $\mathfrak{a}'$, and that

$$\rho_i \parallel \rho'_i$$

defines an isomorphic relation between $\mathfrak{a}$ and $\mathfrak{a}'$. Combining this result with the earlier one, according to which modules of the same degree are isomorphic only when they belong to the same class, makes clear the necessity of the condition stated in Theorem I.

Some simple consequences for the function χ may be drawn from this theorem. First, one sees that for all modules $\mathfrak{a}$ of a class A , χ has the same value, so that the symbol $\chi(\mathfrak{a})$ may conveniently be replaced by

$$\chi(A).$$

Moreover one has:

II. If the systems of invariants of the classes A and A' are identical after the invariants reducing to 1 have been removed, then

$$\chi(A) = \chi(A').$$

§ 5. Relation among the functions φ , χ , ψ .

The functions φ, χ, ψ considered here stand in a very simple relation to one another, made clear by the following argument.

Let A be a class of degree n , let $\mathfrak{a}$ be a module in A , and let R be an r -rowed residue of $\mathfrak{a}$. Then, as a simple consideration shows, the following theorem holds: in order that

$$(\mathfrak{a}, R) = \mathfrak{e} \tag{1}$$

hold, it is necessary and sufficient that for every row ρ of n elements one can find a row σ of r elements satisfying the congruence

$$\sigma R \equiv \rho \pmod{\mathfrak{a}}. \tag{2}$$

Now suppose that (1) is satisfied and that $r = n$. Then all σ for which σR is contained in $\mathfrak{a}$ form a module $\mathfrak{a}'$ of degree n ; all σ satisfying a congruence of the form (2), where ρ is regarded as given, are representatives of a definite residue of $\mathfrak{a}'$; and conversely, for all representatives of a definite residue of $\mathfrak{a}'$, σR represents one and the same residue of $\mathfrak{a}$. Thus one obtains a one-to-one correspondence between the residues of $\mathfrak{a}$ and those of $\mathfrak{a}'$ by making the residue σ of $\mathfrak{a}'$ correspond to the residue σR of $\mathfrak{a}$. This correspondence is, as one easily sees, an isomorphism. It follows that the module $\mathfrak{a}'$ belongs to the class A (§ 4, I). We shall say that the residue R of $\mathfrak{a}$ belongs to the module $\mathfrak{a}'$, and we shall denote the isomorphic relation mediated by R by $[R]$.

If R' is also an n -rowed residue of $\mathfrak{a}$ satisfying condition (1), but distinct from R , then R' may well belong to the same module $\mathfrak{a}'$, but the isomorphic relation $[R']$ is different from $[R]$. Otherwise, for every row σ the congruence

$$\sigma R \equiv \sigma R' \pmod{\mathfrak{a}}$$

would have to hold; and by putting for σ successively the rows of the system

$$E^0 = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix},$$

one would infer the congruence

$$R' \equiv R \pmod{\mathfrak{a}}.$$

On the other hand, it is easy to show that, when R is allowed to run through every n -rowed residue of $\mathfrak{a}$ satisfying (1), the symbol $[R]$ represents exactly once every isomorphic relation possible between $\mathfrak{a}$ and any module $\mathfrak{a}'$ of A . Indeed, suppose such a relation between $\mathfrak{a}$ and $\mathfrak{a}'$ is given. If it assigns to the residues of $\mathfrak{a}'$ represented by the n rows of E^0 the residues $\rho_1, \rho_2, \dots, \rho_n$ of $\mathfrak{a}$, and if R is the n -rowed residue formed from them, then the definition of isomorphism gives that the residue σ of $\mathfrak{a}'$ must correspond to the residue σR of $\mathfrak{a}$. Since every residue of $\mathfrak{a}$ is thus representable in the form σR , it first follows that R satisfies

$$(\mathfrak{a}, R) = \mathfrak{e},$$

and then that the given isomorphic relation is identical with the one denoted by $[R]$.

The result of these considerations is that each of the residues R , whose number is $\varphi(A)$, belongs to a definite one of the $\psi(A)$ modules of the class A , and that to each module $\mathfrak{a}'$ of A there belong $\chi(A)$ residues R , namely as many as there are isomorphic relations between $\mathfrak{a}$ and $\mathfrak{a}'$. Hence one obtains the important relation

$$\text{I.} \quad \varphi(A) = \chi(A) \cdot \psi(A).$$

From this and the formulas (§ 3, II; § 2, XVII) one obtains further

$$\chi(A) = \prod_p \chi(A_p),$$

which of course could also be proved directly.

§ 6. Determination of the functions ψ and χ .

From the theorems § 2, IX; § 4, II; § 5, I it is now easy to express the functions ψ and χ , first of all, as quotients of products of φ -functions.

Let $A = \text{Cl}(a_1 \mid a_2 \mid \dots \mid a_n)$ be any class of degree n and positive norm, and put

$$A' = \text{Cl}\left(1 \mid \frac{a_2}{a_1} \mid \frac{a_3}{a_1} \mid \dots \mid \frac{a_n}{a_1}\right),$$

$$A_i = \text{Cl}\left(\frac{a_{i+1}}{a_i} \mid \frac{a_{i+2}}{a_i} \mid \dots \mid \frac{a_n}{a_i}\right), \quad A'_i = \text{Cl}\left(1 \mid \frac{a_{i+2}}{a_{i+1}} \mid \frac{a_{i+3}}{a_{i+1}} \mid \dots \mid \frac{a_n}{a_{i+1}}\right),$$

so that A_i, A'_i are classes of degree $n - i$. Then

$$A = a_1 A', \quad A_i = \frac{a_{i+1}}{a_i} A'_i,$$

$$\psi(A) = \psi(A'), \quad \psi(A_i) = \psi(A'_i) \quad (\S 2, IX),$$

$$\psi(A') = \frac{\varphi(A')}{\chi(A')}, \quad \psi(A'_i) = \frac{\varphi(A'_i)}{\chi(A'_i)} \quad (\S 5, I),$$

$$\chi(A') = \chi(A_1), \quad \chi(A'_i) = \chi(A_{i+1}) \quad (\S 4, II),$$

$$\chi(A_1) = \frac{\varphi(A_1)}{\psi(A_1)}, \quad \chi(A_{i+1}) = \frac{\varphi(A_{i+1})}{\psi(A_{i+1})} \quad (\S 5, I),$$

and hence

$$\psi(A) = \frac{\varphi(A')}{\varphi(A_1)} \psi(A_1), \quad \psi(A_i) = \frac{\varphi(A'_i)}{\varphi(A_{i+1})} \psi(A_{i+1}).$$

Since $\psi(A_{n-1}) = 1$, these relations give

$$\text{I.} \quad \psi(A) = \frac{\varphi(A')\varphi(A'_1)\cdots\varphi(A'_{n-2})}{\varphi(A_1)\varphi(A_2)\cdots\varphi(A_{n-1})},$$

$$\text{II.} \quad \chi(A) = \frac{\varphi(A)\varphi(A_1)\cdots\varphi(A_{n-1})}{\varphi(A')\varphi(A'_1)\cdots\varphi(A'_{n-2})}.$$

The further calculation, which we shall carry out only for the function ψ , first requires treating the case of a primary class. Thus let

$$\text{Cl}(a_1 \mid a_2 \mid \cdots \mid a_n) = A = A_p = \text{Cl}(p^{r_1} \mid p^{r_2} \mid \cdots \mid p^{r_n})$$

and consequently

$$r_1 \leq r_2 \leq \cdots \leq r_n.$$

Let $e_1, e_2, \dots, e_k$ denote the distinct exponents among the r 's, in their natural order. If e_1 occurs s_1 times, e_2 occurs s_2 times, and so on, e_k occurs s_k times among the exponents r , then

$$s_1 + s_2 + \cdots + s_k = n. \quad (1)$$

Among the differences $r_m - r_l$ with $m > l$, the difference $e_i - e_h$ ($i > h$) occurs $s_i s_h$ times. Hence

$$\sum_{i>h} (e_i - e_h) s_i s_h = \sum_{m>l} r_m - r_l = (1-n)r_1 + (3-n)r_2 + (5-n)r_3 + \cdots + (n-3)r_{n-1} + (n-1)r_n. \quad (2)$$

Now put

$$s_1 + s_2 + \cdots + s_i = t_i \quad (i = 1, 2, \dots, k).$$

From

$$p^{e_1} = a_1 = a_2 = \cdots = a_{t_1}, \quad p^{e_2} = a_{t_1+1} = \cdots = a_{t_2}, \quad \dots, \quad p^{e_k} = a_{t_{k-1}+1} = \cdots = a_{t_k}, \quad (3)$$

it follows that in (I) only the classes

$$A', A'_1, \dots, A'_{k-2} \quad \text{and} \quad A_1, \dots, A_{k-1}$$

occur essentially distinctly. More precisely,

$$\psi(A) = \frac{\varphi(A')\varphi(A'_1)\cdots\varphi(A'_{k-2})}{\varphi(A_1)\varphi(A_2)\cdots\varphi(A_{k-1})}. \quad (4)$$

The classes A', A_i, A'_i have degrees $n, n - t_i, n - t_i$, respectively, while the numbers of their invariants divisible by p are respectively $n - t_1, n - t_i, n - t_{i+1}$. Therefore (§ 3, IIIa)

$$\varphi(A') = (\text{Nm } A')^n \frac{\vartheta(p; n)}{\vartheta(p; t_1)} = (\text{Nm } A')^n \frac{\vartheta(p; n)}{\vartheta(p; s_1)},$$

$$\varphi(A_i) = (\text{Nm } A_i)^{n-t_i} \vartheta(p; n - t_i),$$

$$\varphi(A'_i) = (\text{Nm } A'_i)^{n-t_i} \frac{\vartheta(p; n - t_i)}{\vartheta(p; t_{i+1} - t_i)} = (\text{Nm } A'_i)^{n-t_i} \frac{\vartheta(p; n - t_i)}{\vartheta(p; s_{i+1})}.$$

Thus, by (4),

$$\psi(A) = \frac{(\text{Nm } A')^n (\text{Nm } A'_1)^{n-t_1} \cdots (\text{Nm } A'_{k-2})^{n-t_{k-2}}}{(\text{Nm } A_1)^{n-t_1} (\text{Nm } A_2)^{n-t_2} \cdots (\text{Nm } A_{k-1})^{n-t_{k-1}}} \frac{\vartheta(p; n)}{\vartheta(p; s_1) \vartheta(p; s_2) \cdots \vartheta(p; s_k)}. \quad (5)$$

Now

$$\frac{(\text{Nm } A'_{i-1})^{n-t_{i-1}}}{(\text{Nm } A_i)^{n-t_i}} = \left(\frac{\text{Nm } A'_{i-1}}{\text{Nm } A_i} \right)^{n-t_i} (\text{Nm } A'_{i-1})^{t_i-t_{i-1}},$$

and hence

$$\frac{(\text{Nm } A')^n (\text{Nm } A'_1)^{n-t_1} \cdots (\text{Nm } A'_{k-2})^{n-t_{k-2}}}{(\text{Nm } A_1)^{n-t_1} (\text{Nm } A_2)^{n-t_2} \cdots (\text{Nm } A_{k-1})^{n-t_{k-1}}} = \left(\frac{\text{Nm } A'}{\text{Nm } A_1} \right)^{t_1} \left(\frac{\text{Nm } A'_1}{\text{Nm } A_2} \right)^{t_2} \cdots \left(\frac{\text{Nm } A'_{k-3}}{\text{Nm } A_{k-2}} \right)^{t_{k-2}} (\text{Nm } A'_{k-2})^{t_{k-1}}. \quad (6)$$

Furthermore,

$$\begin{aligned} \frac{\text{Nm } A'_{i-1}}{\text{Nm } A'_i} &= \left(\frac{a_{t_i+1}}{a_{t_i}} \right)^{n-t_i}, & \text{Nm } A'_{k-2} &= \left(\frac{a_n}{a_{t_{k-1}}} \right)^{n-t_{k-1}}, \\ \prod_{i=1}^{k-2} \left(\frac{\text{Nm } A'_{i-1}}{\text{Nm } A'_i} \right)^{t_i} (\text{Nm } A'_{k-2})^{t_{k-1}} &= \prod_{i=1}^{k-1} \left(\frac{a_{t_i+1}}{a_{t_i}} \right)^{(n-t_i)t_i} = \prod_{i=1}^{n-1} \left(\frac{a_{i+1}}{a_i} \right)^{i(n-i)} = \prod_{i=1}^n a_i^{2i-1-n}. \end{aligned} \quad (7)$$

From (5), (6), (7) one now obtains

$$\text{III.} \quad \psi(A) = a_1^{1-n} a_2^{3-n} a_3^{5-n} \cdots a_{n-1}^{n-3} a_n^{n-1} \frac{\vartheta(p; n)}{\vartheta(p; s_1) \vartheta(p; s_2) \cdots \vartheta(p; s_k)}.$$

or, taking (2), (3) into account,

$$\text{IIIa.} \quad \psi(A) = p^{\sum_{i>h} s_i s_h (e_i - e_h)} \frac{\vartheta(p; n)}{\vartheta(p; s_1) \vartheta(p; s_2) \cdots \vartheta(p; s_k)}.$$

Introduce a function $g(x; n, m)^2$ by the equation

$$g(x; n, m) = \frac{(1-x^n)(1-x^{n-1}) \cdots (1-x^{n-m+1})}{(1-x)(1-x^2) \cdots (1-x^m)}, \quad (8)$$

where n is an integer and m a positive integer. Then

$$g(x; n, m) = g(x; n-1, m) + x^{n-m} g(x; n-1, m-1), \quad (9)$$

and this equation also holds for arbitrary integral m if one stipulates that

$$g(x; n, 0) = 1, \quad (10)$$

$$g(x; n, m) = 0 \quad (m < 0). \quad (11)$$

There is also the relation

$$g(x; n, m) = x^{m(n-m)} g\left(\frac{1}{x}; n, m\right). \quad (12)$$

From (9), (10), (11) it follows that $g(x; n, m)$ is, for $n \geq 0$, an integral function of x with only nonnegative integer coefficients. For $n \geq m \geq 0$ the functions g and ϑ satisfy

$$g\left(\frac{1}{x}; n, m\right) = \frac{\vartheta(x; n)}{\vartheta(x; m) \vartheta(x; n-m)}. \quad (13)$$

Writing, for abbreviation,

$$\psi(A) = p^{\sum_{i>h} s_i s_h (e_i - e_h)} \bar{\psi}(A) = a_1^{1-n} a_2^{3-n} \cdots a_n^{n-1} \bar{\psi}(A), \quad (14)$$

²It is the same function denoted by (n, m) in Gauss's paper "Summatio quarundam serierum singularium".

one obtains the following expressions for the functions $\psi, \bar{\psi}$:

$$\bar{\psi}(A) = g\left(\frac{1}{p}; n, s_1\right) g\left(\frac{1}{p}; n - s_1, s_2\right) \cdots g\left(\frac{1}{p}; n - s_1 - \cdots - s_{k-2}, s_{k-1}\right), \quad (15)$$

$$\text{IIIb.} \quad \psi(A) = p^{\sum_{i>h} s_i s_h (e_i - e_h - 1)} g(p; n, s_1) g(p; n - s_1, s_2) \cdots g(p; n - s_1 - \cdots - s_{k-2}, s_{k-1}). \quad (16)$$

This representation shows that $\psi(A)$ is an integral function of p , that the coefficient of the highest power of p is equal to 1, and that the remaining coefficients are nonnegative integers.

For an arbitrary class A of positive norm, § 2, XVII reduces the computation of $\psi(A)$ to the case of a primary class. If one sets

$$\prod_p \bar{\psi}(A_p) = \bar{\psi}(A),$$

then

$$\psi(A) = \prod_p \left((a_1)_p^{1-n} (a_2)_p^{3-n} \cdots (a_n)_p^{n-1} \bar{\psi}(A_p) \right) = a_1^{1-n} a_2^{3-n} \cdots a_n^{n-1} \bar{\psi}(A),$$

so that equation (14) holds in general.

Applying the results obtained to the case $n = 2$, one obtains, for $A = \text{Cl}(a_1 \mid a_2)$,

$$\psi(A) = \frac{a_2}{a_1} \bar{\psi}(A) = \frac{a_2}{a_1} \prod_{p|a_2/a_1} \bar{\psi}(A_p) = \frac{a_2}{a_1} \prod_{p|a_2/a_1} \frac{\vartheta(p; 2)}{(\vartheta(p; 1))^2} = \frac{a_2}{a_1} \prod_{p|a_2/a_1} \left(1 + \frac{1}{p}\right),$$

where the product extends over the primes dividing a_2/a_1 .

Example. For

$$a_1 = 3, \quad a_2 = 90$$

one has

$$\frac{a_2}{a_1} = 30 = 2 \cdot 3 \cdot 5,$$

$$\psi(A) = 30 \cdot \frac{3}{2} \cdot \frac{4}{3} \cdot \frac{6}{5} = 72.$$

For

$$n = 3, \quad A = \text{Cl}(a_1 \mid a_2 \mid a_3),$$

one has

$$\psi(A) = \left(\frac{a_3}{a_1}\right)^2 \bar{\psi}(A) = \left(\frac{a_3}{a_1}\right)^2 \prod_{p|a_3/a_1} \frac{\vartheta(p; 3)}{\vartheta(p; 1)\vartheta(p; 2)} \prod_{\substack{p|a_2/a_1 \\ p|a_3/a_2}} \frac{\vartheta(p; 2)}{\vartheta(p; 1)\vartheta(p; 1)},$$

so

$$\psi(A) = \left(\frac{a_3}{a_1}\right)^2 \prod_{p|a_3/a_1} \left(1 + \frac{1}{p} + \frac{1}{p^2}\right) \prod_{\substack{p|a_2/a_1 \\ p|a_3/a_2}} \left(1 + \frac{1}{p}\right),$$

where the first product extends over the primes dividing a_3/a_1 , and the second over the primes that divide both a_2/a_1 and a_3/a_2 .

Example. For

$$a_1 = 2, \quad a_2 = 210, \quad a_3 = 6930$$

one has

$$\frac{a_3}{a_1} = 3465 = 3^2 \cdot 5 \cdot 7 \cdot 11, \quad \frac{a_2}{a_1} = 105 = 3 \cdot 5 \cdot 7, \quad \frac{a_3}{a_2} = 33 = 3 \cdot 11.$$

$$\psi(A) = (3465)^2 \cdot \frac{13}{9} \cdot \frac{31}{25} \cdot \frac{57}{49} \cdot \frac{133}{121} \cdot \frac{4}{3} = 36661716.$$

§ 7. Introduction of several functions depending on several classes.

The function ψ forms the basis for determining other functions which are important for the theory of modules and which will be treated below. We shall consider only modules and classes of positive norm.

A module $\mathfrak{a}$ of positive norm has, as one easily sees, only finitely many divisors, which (§ 2, III) are distributed among those classes that are divisors of $\text{Cl } \mathfrak{a}$. We now focus on the divisors of $\mathfrak{a}$ which belong to a specified class B . If

$$\mathfrak{b}_1, \mathfrak{b}_2, \dots, \mathfrak{b}_k$$

are these modules and E denotes a unimodular system, then (§ 2, VI)

$$\mathfrak{b}_1 E, \mathfrak{b}_2 E, \dots, \mathfrak{b}_k E$$

are all divisors of $\mathfrak{a}E$ within the class B . It follows that the number of divisors of $\mathfrak{a}$ within B depends only on the classes $A = \text{Cl } \mathfrak{a}$ and B . We denote this number by

$$t(A | B).$$

Similarly, the number of multiples which a module $\mathfrak{b}$ of class B possesses within the class A is a function solely of the classes A and B . Let it be denoted by

$$v(A | B).$$

The number of pairs of modules $\mathfrak{a}_i, \mathfrak{b}_k$ satisfying

$$\text{Cl } \mathfrak{a}_i = A, \quad \text{Cl } \mathfrak{b}_k = B, \quad (\mathfrak{a}_i, \mathfrak{b}_k) = \mathfrak{b}_k$$

is represented by each of the expressions $\psi(A)t(A | B)$ and $\psi(B)v(A | B)$. Thus

$$\text{I.} \quad \psi(A)t(A | B) = \psi(B)v(A | B).$$

Furthermore § 2, XV gives the formulas

$$\text{II.} \quad t(A | B) = \prod_p t(A_p | B_p),$$

$$\text{III.} \quad v(A | B) = \prod_p v(A_p | B_p).$$

From § 2, III we know that $t(A | B)$ and $v(A | B)$ are nonzero or zero according as A is or is not divisible by B . Finally, for every class A one has (§ 2, IV)

$$\text{IV.} \quad t(A | A) = v(A | A) = 1.$$

Other class-functions are reached by the following considerations.

If the modules $\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s$, which need not all be distinct, are divisors of a module $\mathfrak{c}$, then their least common multiple $[\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s]$ is also a divisor of $\mathfrak{c}$. One may therefore pose the problem of determining, for a given $\mathfrak{c}$, s modules $\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s$ such that

$$[\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s] = \mathfrak{c}.$$

This problem is of course always solvable, and in general in many ways. It is different, however, if one imposes the condition that each module $\mathfrak{a}_i$ shall belong to a specified given class A_i .

From § 2, VII follows the theorem: if for some module $\mathfrak{c}$ of class Γ it is possible to represent it as the least common multiple of s modules $\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s$ belonging respectively to the classes $A_1, A_2, \dots, A_s$, then the same is possible for every module of the class Γ . More generally, the number of systems of s modules $\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s$ satisfying

$$\text{Cl } \mathfrak{a}_1 = A_1, \quad \text{Cl } \mathfrak{a}_2 = A_2, \quad \dots \quad \text{Cl } \mathfrak{a}_s = A_s; \quad [\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s] = \mathfrak{c}$$

is the same for every module $\mathfrak{c}$ of class Γ . We denote this number by

$$\bar{\gamma}(\Gamma; A_1, A_2, \dots, A_s).$$

It is a function of the classes $\Gamma, A_1, \dots, A_s$, symmetric in $A_1, \dots, A_s$. The question whether a module $\mathfrak{c}$ of class Γ can be represented as the least common multiple of s modules $\mathfrak{a}_1, \dots, \mathfrak{a}_s$ belonging respectively to the classes $A_1, \dots, A_s$ is therefore exactly the question whether $\bar{\gamma}(\Gamma; A_1, \dots, A_s) > 0$. In any given case this can of course be decided by a finite number of trials. The problem, however, is whether one can state general, readily surveyable relations whose existence among the classes $\Gamma, A_1, \dots, A_s$ is the necessary and sufficient condition for the nonvanishing of the function $\bar{\gamma}(\Gamma; A_1, \dots, A_s)$.

First one sees that $\bar{\gamma}(\Gamma; A_1, \dots, A_s)$ can be nonzero only when Γ is divisible by each of the classes $A_1, \dots, A_s$. Thus, if the classes $A_1, \dots, A_s$ are regarded as given, their least common multiple $\Gamma' = [A_1, \dots, A_s]$ is, in a sense, the minimum among the classes Γ for which $\bar{\gamma}(\Gamma; A_1, \dots, A_s) > 0$: first, $\bar{\gamma}(\Gamma'; A_1, \dots, A_s) > 0$, since the principal module of Γ' is the least common multiple of the principal modules of $A_1, \dots, A_s$; and second, every class Γ satisfying $\bar{\gamma}(\Gamma; A_1, \dots, A_s) > 0$ is a multiple of Γ' . A further investigation shows that among the classes Γ under discussion there also exists a maximum, namely one of them, call it Γ'' , that is divisible by all the others. The construction of the class Γ'' will be given shortly.

Entirely analogous results are obtained in the investigation of the greatest common divisor of modules. We denote by

$$\bar{\delta}(\Delta; A_1, \dots, A_s)$$

the number of systems of s modules $\mathfrak{a}_1, \dots, \mathfrak{a}_s$ respectively taken from the classes $A_1, \dots, A_s$ and having a given module $\mathfrak{b}$ of class Δ as their greatest common divisor. If the classes $A_1, \dots, A_s$ are regarded as given, then all classes Δ for which $\bar{\delta}(\Delta; A_1, \dots, A_s) > 0$ lie between a maximum Δ' and a minimum Δ'' , so that

$$\bar{\delta}(\Delta'; A_1, \dots, A_s) > 0, \quad \bar{\delta}(\Delta''; A_1, \dots, A_s) > 0,$$

and every other class of the kind considered is a divisor of Δ' and a multiple of Δ'' . Here Δ' is plainly the greatest common divisor of the classes $A_1, \dots, A_s$.

To obtain the classes Γ'', Δ'' , proceed as follows. Let

$$A_i = \text{Cl}(a_{i1} \mid a_{i2} \mid \dots \mid a_{in}) \quad (i = 1, 2, \dots, s).$$

From all $s \cdot n$ invariants

$$a_{11}, \dots, a_{1n}, \dots, a_{s1}, \dots, a_{sn}$$

of the classes $A_1, \dots, A_s$, form a diagonal system K and determine its class K . This has degree $s \cdot n$ and is plainly independent of the order of the classes $A_1, \dots, A_s$. The first n invariants of K form the invariant system of a class of degree n , which we call the “limiting divisor” of the classes $A_1, \dots, A_s$ and denote by

$$[A_1, \dots, A_s].$$

Similarly, the last n invariants of K form the invariant system of a class of degree n . This class is called the “limiting multiple” of the classes $A_1, \dots, A_s$ and is denoted by

$$\llbracket A_1, \dots, A_s \rrbracket.$$

It is the maximum mentioned above for the classes Γ for which $\bar{\gamma}(\Gamma; A_1, \dots, A_s) > 0$; likewise, the limiting divisor of $A_1, \dots, A_s$ is the minimum of the classes Δ for which $\bar{\delta}(\Delta; A_1, \dots, A_s) > 0$. It is not hard to prove this without entering into a general determination of the functions $\bar{\gamma}$ and $\bar{\delta}$. But the question now arises whether conversely, for all classes Γ lying between $\llbracket A_1, \dots, A_s \rrbracket$ and $[A_1, \dots, A_s]$, and for all classes Δ lying between $(A_1, \dots, A_s)$ and $(\lceil A_1, \dots, A_s \rceil)$, the functions $\bar{\gamma}(\Gamma; A_1, \dots, A_s)$ and $\bar{\delta}(\Delta; A_1, \dots, A_s)$ are positive. I was able to decide this question only by a detailed study of the functions $\bar{\gamma}$ and $\bar{\delta}$; it turned out that the question just raised must be answered affirmatively. Thus one obtains the following double theorem, which is also of interest especially for the theory of groups with commuting elements:

V. a. In order that a module $\mathfrak{c}$ of class Γ can be represented as the least common multiple of s modules $\mathfrak{a}_1, \dots, \mathfrak{a}_s$ belonging respectively to the classes $A_1, \dots, A_s$, it is necessary and sufficient that Γ lie between the limiting multiple and the least common multiple of the classes $A_1, \dots, A_s$.

b. In order that a module $\mathfrak{b}$ of class Δ can be represented as the greatest common divisor of s modules $\mathfrak{a}_1, \dots, \mathfrak{a}_s$ belonging respectively to the classes $A_1, \dots, A_s$, it is necessary and sufficient that Δ lie between the greatest common divisor and the limiting divisor of the classes $A_1, \dots, A_s$.

In what follows, the proof of this theorem is completely supplied by the investigation of the functions $\bar{\gamma}$ and $\bar{\delta}$.

From § 2, XVI follow the relations

$$\text{VI.} \quad \bar{\gamma}(\Gamma; A_1, \dots, A_s) = \prod_p \bar{\gamma}(\Gamma_p; (A_1)_p, \dots, (A_s)_p),$$

$$\bar{\delta}(\Delta; A_1, \dots, A_s) = \prod_p \bar{\delta}(\Delta_p; (A_1)_p, \dots, (A_s)_p).$$

It is also easy to see from this that, in order to prove V, it is necessary only to prove it for primary classes.

§ 8. The functions t and v expressed by the function ψ .

We now turn to a closer examination of the functions $t(A \mid B)$ and $v(A \mid B)$, and first show how they can be represented with the help of ψ -functions. This can be done in various ways.

Let $A = \text{Cl}(a_1 \mid a_2 \mid \dots \mid a_n)$, $B = \text{Cl}(b_1 \mid b_2 \mid \dots \mid b_n)$, let $\mathfrak{a}$ be an arbitrary module of A , and finally let

$$\mathfrak{b}_1, \mathfrak{b}_2, \dots, \mathfrak{b}_{\psi(B)} \tag{1}$$

be all modules of B . The module $\mathfrak{a}$ is divisible by $a_1 \mathfrak{e}$ (§ 2, X); hence, if $\mathfrak{a}$ is divisible by $\mathfrak{b}_k$, it is also divisible by $[a_1 \mathfrak{e}, \mathfrak{b}_k]$, and conversely. The modules

$$[a_1 \mathfrak{e}, \mathfrak{b}_1], [a_1 \mathfrak{e}, \mathfrak{b}_2], \dots, [a_1 \mathfrak{e}, \mathfrak{b}_{\psi(B)}] \tag{2}$$

all belong to the class $[a_1 E, B]$ (§ 2, XI), and a simple consideration shows that every module of $[a_1 E, B]$ occurs in the sequence (2) equally often, namely

$$\frac{\psi(B)}{\psi([a_1 E, B])}$$

times. Since $\mathfrak{a}$ is divisible by $t(A \mid [a_1 E, B])$ modules of the class $[a_1 E, B]$, one obtains:

I. If a_1 is the first invariant of the class A , then

$$t(A \mid B) = \frac{\psi(B)}{\psi([a_1 E, B])} \cdot t(A \mid [a_1 E, B]).$$

Furthermore:

II. If the classes

$$A = \text{Cl}(a_1 \mid a_2 \mid \cdots \mid a_n) \quad \text{and} \quad B = \text{Cl}(b_1 \mid b_2 \mid \cdots \mid b_n)$$

have the same first invariant $a_1 = b_1$, and if

$$A' = \text{Cl}(a_2 \mid \cdots \mid a_n), \quad B' = \text{Cl}(b_2 \mid \cdots \mid b_n)$$

is put, then

$$t(A \mid B) = t(A' \mid B').$$

Proof. Let A be the principal system of A , let α be its first row, let $\text{Md } A = \mathfrak{a}$, let $\mathfrak{a}'$ be the principal module of A' , and let $\mathfrak{b} = \text{Md } B$ be any divisor of $\mathfrak{a}$ within the class B . If

$$B = \begin{pmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & & \vdots \\ b_{n1} & \cdots & b_{nn} \end{pmatrix}$$

and if $\beta_1, \beta_2, \dots, \beta_n$ denote the rows of B , then first $\mathfrak{b} = \text{Md}(\beta_1, \dots, \beta_n)$; and since $\mathfrak{a}$ is divisible by $\mathfrak{b}$, so that the row α belongs to the module $\mathfrak{b}$, one has

$$\mathfrak{b} = \text{Md}(\alpha, \beta_1, \dots, \beta_n).$$

From $\text{Cl } B = B$ and $a_1 = b_1$ it follows that all elements b_{i1} are divisible by a_1 . Put $b_{i1} = m_i a_1$ ($i = 1, \dots, n$) and $\beta_i - m_i \alpha = \gamma_i$. Then

$$\mathfrak{b} = \text{Md}(\alpha, \gamma_1, \dots, \gamma_n).$$

In γ_i the first element is 0; let γ'_i denote the row of $n - 1$ elements left after deleting this zero. Then $\mathfrak{b}' = \text{Md}(\gamma'_1, \dots, \gamma'_n)$ is a module of degree $n - 1$. If

$$B' = \begin{pmatrix} b'_{11} & \cdots & b'_{1,n-1} \\ \vdots & & \vdots \\ b'_{n-1,1} & \cdots & b'_{n-1,n-1} \end{pmatrix} \quad (3)$$

is any square basis of this module, then

$$\bar{B} = \begin{pmatrix} a_1 & 0 & \cdots & 0 \\ 0 & b'_{11} & \cdots & b'_{1,n-1} \\ \vdots & \vdots & & \vdots \\ 0 & b'_{n-1,1} & \cdots & b'_{n-1,n-1} \end{pmatrix} \quad (4)$$

is a basis of $\mathfrak{b}$. From $\text{Cl } \bar{B} = B$ it follows easily that $\text{Cl } \mathfrak{b}' = \text{Cl } B' = B'$, and from the divisibility of $\mathfrak{a}$ by $\mathfrak{b}$ it follows that $\mathfrak{a}'$ is divisible by $\mathfrak{b}'$. Conversely, the following is easy to see: if one has two square systems $B', \bar{B}$ represented as in (3) and (4), and if $\text{Md } B' = \mathfrak{b}'$ is a divisor of $\mathfrak{a}'$ with $\text{Cl } B' = B'$, then $\text{Md } \bar{B} = \mathfrak{b}$ is a divisor of $\mathfrak{a}$ and $\text{Cl } \bar{B} = B$. Finally, one sees that, in the way described, the module $\mathfrak{b}'$ is uniquely determined by $\mathfrak{b}$, and likewise $\mathfrak{b}$ by $\mathfrak{b}'$. Thus $\mathfrak{a}$ has just as many divisors in the class B as $\mathfrak{a}'$ has in the class B' , that is,

$$t(A \mid B) = t(A' \mid B').$$

Now let $A = \text{Cl}(a_1 \mid a_2 \mid \cdots \mid a_n)$ be divisible by $B = \text{Cl}(b_1 \mid b_2 \mid \cdots \mid b_n)$, and put

$$\begin{cases} \text{Cl}(a_{i+1} \mid a_{i+2} \mid \cdots \mid a_n) = A_i, \\ \text{Cl}([a_i, b_{i+1}] \mid [a_i, b_{i+2}] \mid \cdots \mid [a_i, b_n]) = B_i, \quad (i = 0, 1, \dots, n-1). \\ \text{Cl}([a_{i+1}, b_{i+1}] \mid [a_{i+1}, b_{i+2}] \mid \cdots \mid [a_{i+1}, b_n]) = \bar{B}_i, \end{cases} \quad (5)$$

Then

$$\bar{B}_i = [a_{i+1}E, B_i]$$

and hence by Theorem I

$$t(A_i | B_i) = \frac{\psi(B_i)}{\psi(\bar{B}_i)} t(A_i | \bar{B}_i).$$

Moreover $[a_{i+1}, b_{i+1}] = a_{i+1}$, whence by Theorem II

$$t(A_i | \bar{B}_i) = t(A_{i+1} | B_{i+1}).$$

Finally, taking into account that $t(A_{n-1} | B_{n-1}) = 1$, one obtains the formula

$$\text{III.} \quad t(A | B) = \frac{\psi(B)\psi(B_1) \cdots \psi(B_{n-2})}{\psi(\bar{B})\psi(\bar{B}_1) \cdots \psi(\bar{B}_{n-2})}.$$

The function $v(A | B)$ can also be represented as a quotient of products of ψ -functions by using two theorems that form a dual analogue of I and II. The same result can also be obtained by applying the formulas § 7, I and the formula just found for $t(A | B)$. One then obtains

$$\text{IV.} \quad v(A | B) = \frac{\psi(A)\psi(B_1) \cdots \psi(B_{n-2})}{\psi(\bar{B})\psi(\bar{B}_1) \cdots \psi(\bar{B}_{n-2})}.$$

The further calculation of the functions $t(A | B)$ and $v(A | B)$, in which we may restrict ourselves to primary classes, will be carried out only later.

§ 9. Representation of the function $\bar{\gamma}$ by t -functions and of $\bar{\delta}$ by v -functions.

Having shown in § 8 how the functions t and v can be expressed by ψ -functions, we now show how the functions $\bar{\gamma}$ and $\bar{\delta}$ can be represented with the aid of t and v .